

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior backend engineer with 6+ years shipping **secure, reliable, scalable production systems** for fully remote product teams across the US, Singapore and Uganda. Led a **zero-downtime migration of 20+ production applications** from Parse/MongoDB to Firebase/GCP, reporting to the CTO, with a live two-way sync between datastores during cutover. Builds **agentic workflows** end to end — spec, plan, execute, verify — including an AI pipeline on Claude Code that verifies every input in a headless browser before any LLM step runs. Works async by default and picks up new stacks fast: joined MTailor as a contractor on an unfamiliar platform and became its migration lead.

CORE COMPETENCIES

Backend Systems & API Design

Agentic Workflows (spec → plan → execute → verify)

AI Fluency (Claude Code, OpenAI)

Verification Loops & Guardrails

Zero-Downtime Migrations

Event-Driven Services (Pub/Sub, Kafka)

Docker & Kubernetes

Async Remote Collaboration

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote · e-commerce

- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Implemented **real-time two-way sync** between MongoDB and Firestore in Node.js on Google **Pub/Sub**, keeping two live production systems consistent through the cutover.
- Moved all services off AWS, saving **\$5,000/month**; migrated file storage from Amazon S3 to GCS via a Python script.
- Shipped product features end to end: Express Shipping (**+\$40 revenue per order**) and a 3D visualisation feature (video overlay + ffmpeg) that increased buyer conversion; migrated a Webflow site from Parse to Firebase.

MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Built the migration script that read from MongoDB, ran complex processing logic, and wrote to Firestore.
- Wrote documentation for fellow engineers on the new Firebase SDKs (Node.js and Python); trained the Ops team on the dashboard and backup procedures.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote · fintech PWA

- Led optimisation of a PWA delivering real-time stock prices; built ad-hoc jobs over **2M+ Firestore records** to keep customer prices current from Morningstar APIs.
- Extended backend APIs in TypeScript on Firebase Cloud Functions and Express.js (Heroku); added responsive React/Tailwind pages.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote · agency

- Built backends for DeFi applications in Node.js with Web3; delivered React UIs from Figma designs under tight timelines, reporting to the CTO.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote · legal tech

- Led a team of 4 developers building legal-tech systems for NGOs; architected a microservices backend for FIDA Uganda's case management app on **Apache Kafka, Docker and Kubernetes**.
- Built a USSD service for legal-aid providers communicating with a Node.js backend via **gRPC**; delivered a paralegal database for LASPNET and a React Native app with Firebase push notifications.

PROJECTS

Agentic Evaluation Pipeline agentic workflow

Extended an open-source agentic pipeline on Claude Code that scans job portals, verifies each posting live in a headless browser before any LLM scoring, deduplicates sources and generates tailored documents. Human stays on every irreversible step; 200+ evaluations with verification-before-inference as the core design rule.

Claude Code · Playwright · Node.js · YAML

Healthcare WhatsApp Assistant LLM in production

Conversational assistant that reads prescription photos and responds to patients in WhatsApp: OpenAI for dialogue, Google Vision for OCR, behind the WhatsApp Business API.

OpenAI API · Google Cloud Vision · WhatsApp Business API · Node.js

Real-time Pub/Sub Sync Service

Two-way MongoDB↔Firestore sync running under parallel production traffic across 20+ applications during a zero-downtime migration.

Node.js · Google Pub/Sub · MongoDB · Firestore

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Languages: JavaScript, TypeScript, Python, SQL, Go (intermediate), Rust (intermediate)

Backend: Node.js, Express.js, REST APIs, gRPC, Microservices, Firebase Cloud Functions

AI / Automation: Claude Code agentic workflows, OpenAI API, Google Cloud Vision, Playwright

Data & Messaging: MongoDB, Firestore, Google Pub/Sub, Apache Kafka, NATS Streaming

Cloud / Infra: GCP, Firebase, AWS (S3, EC2, EBS — intermediate), Docker, Kubernetes, Git, Linux

Frontend: React, React Native, Tailwind, HTML, CSS **Spoken:** English, Kiswahili